

Pet Health & Wellness Tracker

AI-Assisted Development Portfolio Report

1. Project Description & Idea

App / Game Idea

The Pet Health & Wellness Tracker is a friendly, easy-to-use website designed to help pet owners look after their furry friends. It allows pet parents to keep all their pet's critical details in one single, safe place. You can log daily activities like meals, walks, medications, and bathroom breaks. The app features state-of-the-art interactive charts to visualize weight variations over time and set alerts for upcoming vet appointments or vaccination updates. It acts as a digital diary for your companion.

Mission & Purpose

The purpose of this project is to solve a common problem for pet owners: remembering vaccine dates, medical records, and daily habits. Instead of using messy paper notes or confusing spreadsheets, this app provides a stress-free dashboard so pets remain happy, healthy, and on track with their wellness requirements.

2. AI-Assisted Development Process & Scaffolding

Neural Models & Developer Tools Used

- Code Assistant Models: Google AI Studio, Gemini 3.5 Flash (AI Coding Assistant), Vite, React 19, Tailwind CSS, Lucide Icons, LocalStorage Persistence
- Integrated Scaffold Runtime: React 19 Engine, Tailwind CSS, TypeScript Modules

Initial Direct Prompt

Create a modern, single-page Pet Health and Wellness Tracker in React with custom styling. The app should let users register their pets with name and type, add routine daily logs (such as foods/meals, habits, medications, and bodyweight), render visual trends of their weight logs, and save all input information on the browser's local storage so nothing is lost when the page refreshes.

Prompt Engineering Iterations

- "Let's make the website look very warm, inviting, and friendly. Customize the layout with soft round corners, clean borders, and pet-friendly color accents like soft brown or sky blue."
- "Add an interactive reminder system where owners can add upcoming vaccination alerts and check them off when done."
- "Introduce an expert pet guide section powered by advice models to show general daily health tips based on pet types (Dogs vs. Cats)."

3. Incremental Sprint Milestones

[COMPLETED] Phase 1: Blueprint: App Shell & UI Design

Crafted the initial friendly interface and established sections for Pet Profiles and Daily Logs.

Prompt Key: "Create a simple React screen for logging pet details with a warm green and beige layout."

[COMPLETED] Phase 2: Data Store: Local Cache Integration

Implemented local storage code blocks to secure client entries and stop state loss when reloading.

Prompt Key: "Write React state logic to save and load pet inputs into the browser's LocalStorage."

[COMPLETED] Phase 3: Chart Trends: Weight Visualization

Constructed standard responsive charts to plot weight entries and added checklist alerts.

Prompt Key: "Build a friendly visual trend graph to represent weight changes for dog profiles."

4. Challenges Met & Applied Solutions

What Worked Perfectly

Gemini was incredibly helpful. It generated the exact list structures and styled the inputs beautifully on the first go. The browser local storage logic was automatically handled, so pet profiles were successfully saved and reloaded immediately.

Did Not Work / Constraints Faced

At the start, adding multiple pets caused the dynamic charts to overlap or break because the chart component didn't know which pet's weight history was currently active.

Bugs Profile Checked

We noticed a bug where entering a decimal value for a pet's weight would convert it into a string text. This caused the weight trend graph to look empty or display weird, flat lines.

Interactive Solutions Applied

We fixed this in React by wrapping the input value in a standard Javascript 'parseFloat()' function before storing it. This made sure weight math values remained pure numbers for the graph rendering.

5. Features & Real-World Utility

Final Working Features

- Pet Profile Registration (Set name, breed, weight, and age)
- Routine Daily Care Tracker (Track meals, walks, meds, and mood)
- Dynamic Weight Progress Estimator & visual chart logs
- Interactive Vaccination & Appointment Checklist alert drawer
- Quick Pet Care Advice Cards showing basic rules

Real-World Problem Solving Value

Yes, this project directly solves a real-world problem. Many pet parents struggle to track recurring medical regimens, weight charts, or coordinate care schedules between household members. This app simplifies caregiver coordination, ensuring pets never miss critical medications.

6. Evaluation KPI Scores

- Design Friendliness - 95% Score

Cutesy fonts, high contrast text fields, and soft shapes.

- **Feature Usefulness - 91% Score**

Addresses daily pet routines and medical tracking.

- **No-Code Feasibility - 100% Score**

Easy for students without computer science training to prompt.